In this Learning Module you will learn how to find equivalent forms of rational numbers including whole numbers, fractions and decimals. In order to do this, you will first need to understand what fractions represent and how to convert (or change) whole numbers to fractions and then to convert fractions to decimals. Read the definitions below.

Whole Number: A number that does not contain a fraction or a decimal.

Fraction: Less than one. A fraction is always written as part over whole.

An example is 1/2 or 3/4. In the fraction 1/2, 1 means 1 part, and 2 means 2 parts in the whole. So the fraction 1/2 means 1 part out of a total of 2 parts.

In the fraction 3/4, the 3 means 3 parts and the 4 means 4 parts in the whole. So the fraction 3/4 means 3 parts out of a total of 4 parts.

Definitions

Read the information and trace the dotted words.

Decimal: A number less than one written with a decimal point to represent a part of a whole.

For example, in the decimal .3, the 3 means 3 parts out of 10. In the decimal .03 the 3 means 3 parts out of 100.

Go back and reread the definitions above, using your marker to circle the words in the definitions that tell you what a whole number, fraction and decimals are.

In the numbers below, the whole numbers have been circled. Use your marker to trace the circles around the whole numbers.

1/2 .3 3/4 .5 7

In the numbers below, the fractions have been circled. Use your marker to trace the circles around the whole numbers

1/2 .3 3/4 9 .5 7

In the numbers below, the decimals have been circled. Use your marker to trace the circles around the decimals.

1/2 .3 3/4 9 .5 7

Definitions

Read the information and trace the dotted words.

Whole Number- A number without a fraction or decimal.

1 2 3 4 5 6

Fraction- A number less than one, written as part over whole.

1/2 1/4 3/4

Decimal- A number less than one, written with a decimal point.

.3 .6 .8 .02 .5 .04

Definitions

Practice identifying whole numbers, fractions, and decimals.

2, 1/8, .4, 8, 3/7, .75, .9, 4/5, 5,
7/8, .5, 9, 4, 3, .58, 8/10, 1,
5/7, .67, 7, 9/12, 6, 5/8, .1

Whole Numbers

Fractions

Decimals

Numbers, Fractions and Decimals

List numbers, fractions and decimals in the correct box.

2, 1/8, .4, 8, 3/7, .75, .9, 4/5, 5,
7/8, .5, 9, 4, 3, .58, 8/10, 1,
5/7, .67, 7, 9/12, 6, 5/8, .1

Whole Numbers

2, 8, 5, 9, 4, 3, 1, 7, 6

Fractions

1/8, 3/7, 4/5, 7/8, 8/10, 5/7, 9/12, 5/8

Decimals

.4, .75, .9, .5, .58, .67, .1

Directions: Put a circle around the numbers that are whole numbers. Put a triangle around the numbers that are fractions. Put a square around the numbers that are decimals.

4 1/6

.9 5/12 .67 6

56 .94 4/5 .4

83 .91 5/18 .65

.5 46 8/12 5

.73 6/13 7/21 3 7

Circle- 4,6,56,83,46,5,3,7
Triangle- 1/6,5/12,4/5,5/18,8/12,6/13,7/21
Square- .9,.67,.94,.4,.91,.65,.5,.73

Numbers, Fractions and Decimals *Follow the directions.*

Converting (Changing) Whole Numbers to a Fraction and a Decimal-

Now that you understand about what whole numbers, fractions and decimals are, you will be able to use that knowledge to find equivalence of these numbers.

Think about a whole number like 4. To convert (change) this to a fraction, place 4 over 1 - 4/1. We are talking about 4 parts when 1 is a whole. To convert (change) this to a decimal, place the decimal point after the 4. Your number would look like this: 4.0.

Some improper fractions (when the numerator is the larger number) can be written as a whole number. (Remember that a numerator is the number on top in a fraction). Think about 25/5- if you divided 25 by 5, you will get a whole number.

$$5\overline{)25} \quad = \quad \frac{25}{5}$$

So, some fractions can be written as a whole number equivalent.

A whole number can be written as a decimal by placing the decimal after the number. Think of 6.00; this means 6 wholes and no parts.

Use your marker to trace changing whole numbers to fractions and decimals and important words in the explanations.

To change a whole number to a fraction, you can place the whole number over 1.

To change a whole number to a decimal, you can add a decimal point after the whole number.

4	=	4/1	=	4.0
7	=	7/1	=	7.0
2	=	2/1	=	2.0
12	=	12/1	=	12.0
23	=	23/1	=	23.0

It is time to practice what you have learned about changing (converting) whole numbers to fractions and decimals.

Circle the letter for the correct answer to each question.

1) Which of the following list shows the correct way to change a whole number to a fraction and then a decimal?

A	5	=	5/2	=	5.2
B	6	=	6/1	=	1.6
C	8	=	1/8	=	8.0
D	9	=	9/1	=	9.0

2) Which of the following list does not show the correct way to change a whole number to a fraction and then a decimal?

A	7	=	7/1	=	7.0
B	3	=	3/1	=	3.0
C	4	=	4/4	=	.4
D	2	=	2/1	=	2.0

It is time to practice what you have learned about changing (converting) whole numbers to fractions and decimals.

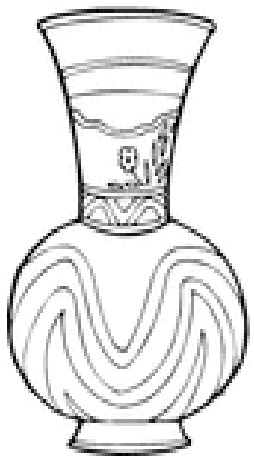
Circle the letter for the correct answer to each question.

1) Which of the following list shows the correct way to change a whole number to a fraction and then a decimal?

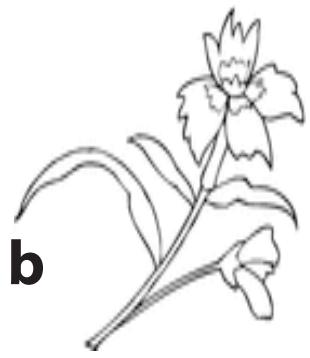
A	5	=	5/2	=	5.2
B	6	=	6/1	=	1.6
C	8	=	1/8	=	8.0
D	9	=	9/1	=	9.0

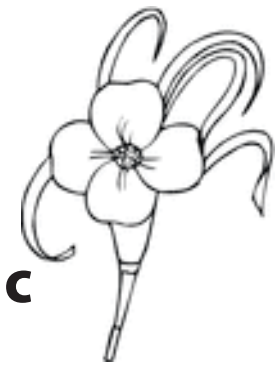
2) Which of the following list does not show the correct way to change a whole number to a fraction and then a decimal?

A	7	=	7/1	=	7.0
B	3	=	3/1	=	3.0
C	4	=	4/4	=	.4
D	2	=	2/1	=	2.0



a

b 

c 

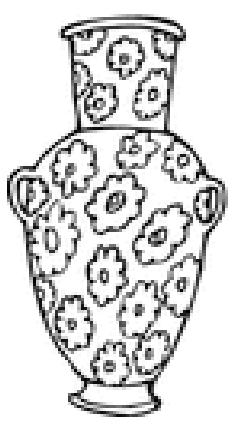
Whole Numbers **5** **.9** **3/4**

a

b

c

Fractions **.2** **1/9** **8**



a

b

c

Decimals **3/8** **.3** **2/3**

Game

Draw the flower in the correct vase.

There are some decimals that can be easily memorized.

1/4 = .25 (Think quarters in a dollar. One quarter =$.25)

1/2 = .50 (Think of half a dollar = $.50)

3/4= .75 (Think of 3 quarters = $.75)

1/3= .33 1/3 2/3 = .66 2/3 2/4=1/2= .50 3/6=1/2= .50

Think about a fraction, say 1/5. This is less than one, so we can not make a whole number out of it. But we can make a decimal. 1/5 can be made into a decimal by dividing 1 by 5.

$$5\overline{)1.00}^{\,.20} = \frac{1}{5}$$

So, fractions can be written as a decimal equivalent.
Now think of 1/8. That is 1 divided by 8.

$$8\overline{)1.00}^{\,.125} = \frac{1}{8}$$

Converting (changing) Fractions to Decimals

Read the information and trace the dotted words.

1/2 = .5 3/4 = .75 1/4 = .25

1/3= .33 1/3 2/3= .66 2/3

3/5= .60 4/8 = 1/2 = .50 9/10 = .90

.111 = 1/9 .166 = 1/6

Practice What You've Learned

Trace the decimal equivalence of each fraction.

Remember equivalent means equal.

1) What is the decimal equivalent of 1/2?

.65 .75 .38 .50

2) What is the decimal equivalent of 3/4?

.75 .4 .8 .70

3) What is the decimal equivalent of 1/4?

.39 .25 .6 .14

4) What is the decimal equivalent of 1/3?

.44 .13 .33 1/3 .56

5) What is the decimal equivalent of 1/5?

.20 .15 .66 2/3 .56

6) What is the decimal equivalent of 9/10?

.85 .78 .28 .90

7) What is the decimal equivalent of 2/3?

.94 .66 2/3 .5 .14 2/7

Equivalence Quiz

Circle the correct answer.

Remember equivalent means equal.

1) What is the decimal equivalent of 1/2?

.65 .75 .38 .50

2) What is the decimal equivalent of 3/4?

.75 .4 .8 .70

3) What is the decimal equivalent of 1/4?

.39 .25 .6 .14

4) What is the decimal equivalent of 1/3?

.44 .13 .33 1/3 .56

5) What is the decimal equivalent of 1/5?

.20 .15 .66 2/3 .56

6) What is the decimal equivalent of 9/10?

.85 .78 .28 .90

7) What is the decimal equivalent of 2/3?

.94 .66 2/3 .5 .14 2/7

Equivalence Quiz *Answers*

1/2	.125
2/3	.25
4/5	.66 2/3
1/4	.50
1/3	.90
9/10	.33 1/3
1/8	.80

Solution- 1/2=.50, 2/3= .66 2/3, 4/5=.80, 1/4=.25, 1/3=.33 1/3, 9/10= .90.

Game

Draw a line from the fraction to the correct decimal.

Now we will practice converting (changing) other fractions to decimals. We have practiced converting easy fractions to decimals. Most of these should be memorized. Now we will work to convert more difficult fractions to decimals. The line in a fraction means to "divide." In the fraction 3/8, 3 is divided by 8.

$$\begin{array}{r} .375 \\ 8\overline{)3.00} \\ \underline{2.40} \\ 0.60 \\ \underline{0.56} \\ 0.04 \end{array}$$

3/8 = .375

More Converting Fractions to Decimals

Read and trace the dotted words.

Now it is your turn. Take these uncommon fractions and convert (change) them to decimals.

1) 4/7

```
     .57 1/7
7 | 4.00
    3.50
    ----
    0.50
    0.49
    ----
    0.01
```

4/7 = .57 1/7

2) 4/9

```
     .44 4/9
9 | 4.00
    3.60
    ----
    0.40
    0.36
    ----
    0.04
```

4/9 = .44 4/9

Converting Uncommon Fractions

Take the uncommonn fractions and convert them to decimals.

1) For the fraction 3/7 you would divide 3 by

_____ .

2) Some easy fractions can be

_____ .

3) Converting fractions means to

_____ them.

4) The line in a fraction means to

_____ .

5) Fractions that can be easily memorized are called

_____ fractions.

Word Bank

divide	**easy**
change	**memorized**
7	

1) For the fraction 3/7 you would divide 3 by

_____ .
7

2) Some easy fractions can be

_____ .
memorized

3) Converting fractions means to

_____ them.
change

4) The line in a fraction means to

_____ .
divide

5) Fractions that can be easily memorized are called

_____ fractions.
easy

Word Bank

divide	**easy**
change	**memorized**
7	

C	Z	E	Q	B	E	J	U	L	D	V	M	M	X	W
U	D	U	U	D	J	U	A	I	W	H	E	V	S	P
W	X	G	I	C	Z	A	F	N	X	H	M	A	L	C
H	P	V	I	S	O	F	X	E	P	Y	O	M	E	D
P	I	U	L	F	I	N	G	R	U	M	R	P	J	U
D	L	C	A	C	H	O	V	Z	S	O	I	D	K	L
T	V	A	U	U	J	Y	S	E	P	L	Z	S	T	L
T	K	L	N	U	W	P	P	F	R	W	E	G	Y	A
I	T	I	L	O	S	H	V	G	V	T	D	M	V	M
O	D	M	Q	G	M	A	E	Z	U	E	I	M	G	I
C	H	A	N	G	E	M	G	A	X	A	S	N	T	C
Z	G	R	C	M	K	M	O	C	S	O	H	V	G	E
M	M	J	G	H	V	G	F	C	G	Y	Y	E	U	D
P	N	O	I	T	C	A	R	F	N	R	B	K	D	D

Word Bank

change	**converting**	**decimal**
difficult	**divide**	**easy**
fraction	**line**	**memorized**

Game

Complete the Word Search.

C	Z	E	Q	B	E	J	U	L	D	V	M	M	X	W
U	D	U	U	D	J	U	A	I	W	H	E	V	S	P
W	X	G	I	C	Z	A	F	N	X	H	M	A	L	C
H	P	V	I	S	O	F	X	E	P	Y	O	M	E	D
P	I	U	L	F	I	N	G	R	U	M	R	P	J	U
D	L	C	A	C	H	O	V	Z	S	O	I	D	K	L
T	V	A	U	U	J	Y	S	E	P	L	Z	S	T	L
T	K	L	N	U	W	P	P	F	R	W	E	G	Y	A
I	T	I	L	O	S	H	V	G	V	T	D	M	V	M
O	D	M	Q	G	M	A	E	Z	U	E	I	M	G	I
C	H	A	N	G	E	M	G	A	X	A	S	N	T	C
Z	G	R	C	M	K	M	O	C	S	O	H	V	G	E
M	M	J	G	H	V	G	F	C	G	Y	Y	E	U	D
P	N	O	I	T	C	A	R	F	N	R	B	K	D	D

Word Bank

change	**converting**	**decimal**
difficult	**divide**	**easy**
fraction	**line**	**memorized**

It is easy to convert (change) decimals to fractions. You just have to read decimals and say the decimal out loud. The decimal .9 is said 9 tenths. This can easily be put into a fraction 9/10. The tenths, hundredths, and thousandths are easily put into fractions by saying the decimal out loud.

The decimal .09 can easily be put into the fraction 9/100.

The decimal .009 can easily be put into a fraction, also- 9/1000.

When you have the decimal .36 and you say it out loud, you will have the fraction 36/100.

Decimal to Fraction

Read the information and trace the dotted words.

1) The decimal .7 can be written as the fraction _____.
(HINT- Say the decimal out loud.)

2) The decimal .84 can be written as the fraction _____.
(HINT- Say the decimal out loud.)

3) The decimal .95 can be written as the fraction _____.
(HINT- Say the decimal out loud.)

4) The decimal .125 can be written as the fraction _____.
(HINT- Say the decimal out loud.)

Once you learn how to read a decimal, you can always write it as a fraction. All you have to do is to say the decimal out loud.

Solution- 1) 7/10 2) 84/100 3) 95/100 4) 125/1000

Decimal to Fraction

Write the decimals as fractions.

1) It is easy to _____ (change) decimals to fractions.

2) The decimal .9 is read _____.

3) The decimal .9 can be converted to the fraction _____.

4) The _____, _____, and _____ are easily put into fractions by saying the _____ out loud.

5) You read the decimal .89 as _____.

6) The decimal .89 can be written as the fraction _____.

Decimal to Fraction

Fill in the blanks.

1) It is easy to _____ (change) decimals to
convert

fractions.

2) The decimal .9 is read _____.
9 tenths

3) The decimal .9 can be converted to the fraction _____.
9/10

4) The _____, _____,
tenths **hundredths**

and _____
thousandths

are easily put into fractions by saying the _____
decimals

out loud.

5) You read the decimal .89 as _____.
89 hundredths

6) The decimal .89 can be written as the fraction _____.
89/100

1) You can easily terconv a decimal into a fraction by saying the decimal out loud.

2) You can read the decimal .7 by saying 7 sentth.

3) The fraction 4/10 is read rouf tenths.

4) Decimals in htsent, redhundths, and touhhsandst can be easily changed (converted) to fractions by saying the decimals out loud.

5) You can read the decimal .15 by saying 15 hundthsred.

6) You can read the rafciont 9/10 by saying 9 tenths.

7) The decimal .58 is dera 58 hundredths.

Solution- 1. Convert 2. Tenths 3. Four 4. Tenths, hundredths, thousandths 5. Hundredths 6. Fraction 7. read

Game

Rearrange the letters in the underlined word to make the correct word.

Now that you have learned how easy it is to convert (change) decimals to fractions, let talk about changing fractions to decimals. As you remember it is really easy to change (convert) decimals into fractions because they are always written in tenths, hundredths and thousandths. If we can convert (change) our fractions to tenths, hundredths or thousandths, we can easily convert (change) them to decimals. We can convert (change) fractions by multiplying the numerator and the denominator by the same number. We want our denominator to be a 10, 100, or 1000.

Fractions to Decimals

Read and trace the dotted words.

If the denominator is 25, you can make that 100 by multiplying it by 4.

So, 3/25 can be converted by multiplying 3 and 25 by 4.

$$\frac{3}{25} \times \frac{4}{4} = \frac{12}{100}$$

This fraction would be written as the decimal .12.

If the denominator is 5, you can make that 10 by multiplying it by 2.

$$\frac{4}{5} \times \frac{2}{2} = \frac{8}{10}$$

This fraction would be written as the decimal .8.

1) If the denominator is 50, what number would you multiply 50 by to make it 100? After you figure this out, don't forget to multiply the numerator by this number also.

2) If the denominator is 20, what number would you multiply 20 by to make it 100?

Solution 1. 2 2. 5

1.) If I want to convert (change) 4/5 to a decimal, I would multiply both the denominator and the numerator by which number?

3, 4, 10, 1, 2

2) If I want to change (convert) 15/25 to a decimal, I would multiply both denominator and the numerator by the same or different number.

3) If I want to change (convert) 4/20 to a decimal, I would multiply both the denominator and the numerator by which number?

5, 10, 2, 1, 3

4) When converting (changing) a fraction to a decimal we check to see if we can multiply the denominator by a number to make the denominator a

_____, _____, or _____.

5) If the denominator can not be multiplied by another number to make it 10, 100, or 1000; we can or can not use this method of switching from fractions to decimals.

6) If I want to change 1/ 4 to a decimal, I would multiply the denominator and numerator by _____.

1.) If I want to convert (change) 4/5 to a decimal, I would multiply both the denominator and the numerator by which number?

3, 4, 10, 1, (2)

2) If I want to change (convert) 15/25 to a decimal, I would multiply both denominator and the numerator by the (same) or different number.

3) If I want to change (convert) 4/20 to a decimal, I would multiply both the denominator and the numerator by which number?

(5), 10, 2, 1, 3

4) When converting (changing) a fraction to a decimal we check to see if we can multiply the denominator by a number to make the denominator a

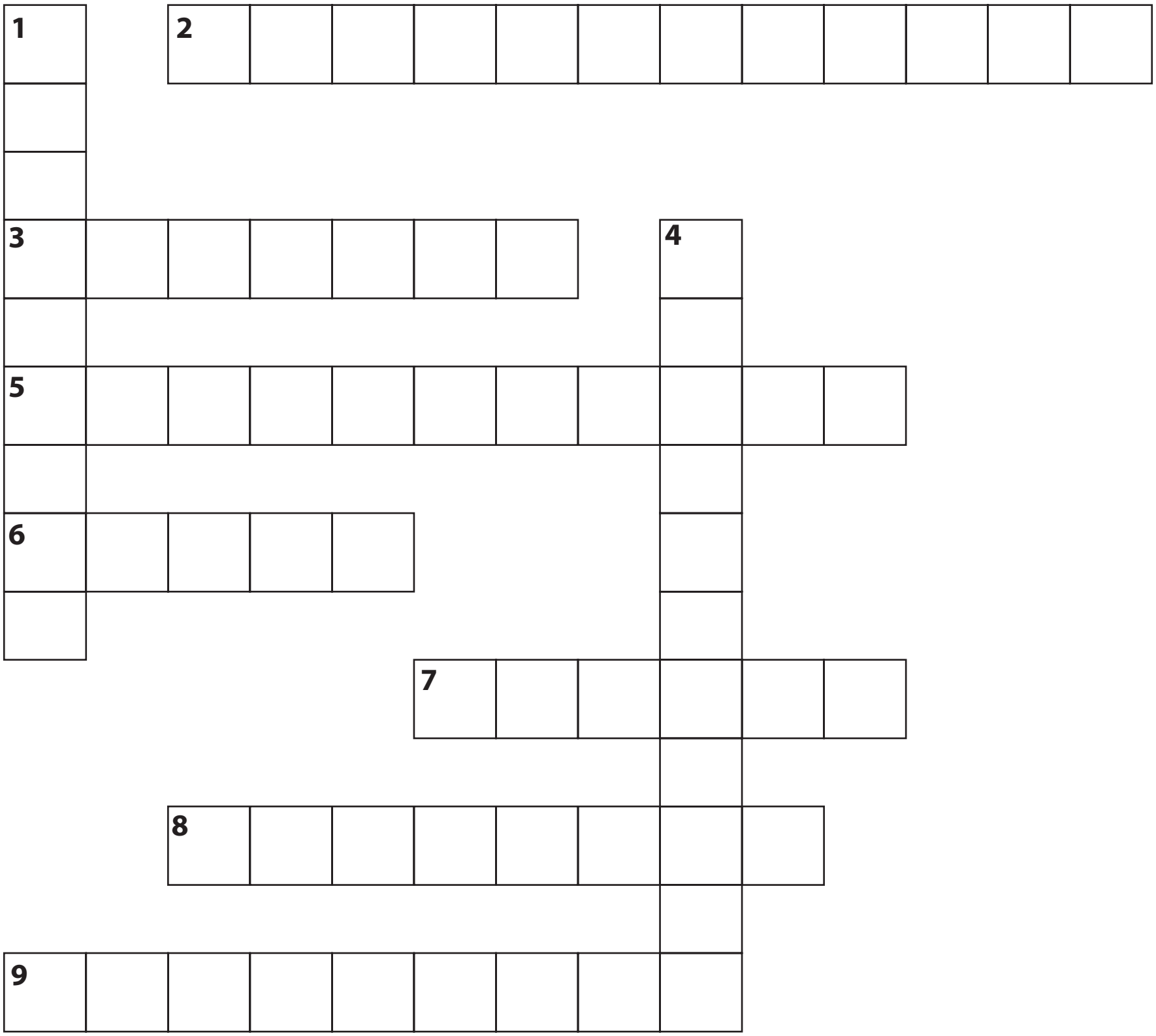
_____, _____, or _____.

10 **100** **1000**

5) If the denominator can not be multiplied by another number to make it 10, 100, or 1000; we can or (can not) use this method of switching from fractions to decimals.

6) If I want to change 1/ 4 to a decimal, I would multiply the denominator and numerator by _____.

25



ACROSS

2. The dot in a number that means less than one.
3. A number written with a decimal point.
5. Equals
6. Part of a whole number (written in the first space after a decimal point).
7. Convert
8. Changes
9. The top number of a fraction.

DOWN

1. Part of a whole number (written in the second space after a decimal point).
4. The bottom number of a fraction.

Game

Fill in the squares.

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